Spectral-Domain PMCHWT-EFIE Solutions of 3-D Electromagnetic Scattering From Multilayered Concentric Dielectric–Metal Spheres

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***Abstract*—Previous research has demonstrated that the spectral-domain integral equation (IE) method significantly outperforms the spatial-domain method in terms of computational cost for calculating electromagnetic (EM) scattering from 3-D smooth homogeneous objects, such as spheres. This superiority arises from the lower sampling density (SD) required by the spectral-domain method, as dictated by the Nyquist theorem. Building on prior research, this study delves into the computation of EM scattering from multilayered concentric dielectric spheres encasing a metal core, a configuration common in modern EM device designs. Multiple spectral-domain Poggio–Miller–Chang–Harrington–Wu–Tsai (PMCHWT) equations and the electric field IE (EFIE) are combined to address both the EM self-coupling within each dielectric or metal spherical surface and the mutual coupling between adjacent surfaces. The final spectral-domain impedance matrix is assembled by forcing the total number of unknown coefficients in the vector spherical harmonics (VSHs) series, used to expand the unknown equivalent surface current densities, to match the number of discretized sampling points on all spherical surfaces. The scattered EM fields at the receiver array are directly computed by multiplying the spectral-domain radiation matrix with the resolved VSH coefficient vector of the surface current densities. Three numerical examples are provided to validate the accuracy and efficiency of the proposed multilayered spectral-domain IE method by comparing its results with those obtained using the Mie series, the traditional spatial-domain IE method, and two well-known commercial software packages. Finally, several typical designs of invisibility cloaks employing spherical shells and subjected to various excitation sources are presented to showcase the utility of the proposed multilayered spectral-domain method.**

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***Index Terms*—Electromagnetic (EM) scattering, integral equation (IE), multilayered spheres, spectral-domain method.**

I. Introduction

THE multilayered spherical structures are ubiquitous in electromagnetic (EM) engineering, including applications, such as radome shielding [1], millimeter-wave lenses [2], resonator design [3], EM invisible cloaking [4], and human head modeling [5]. Notably, the EM scattering from multilayered dielectric spheres with a metal core has been extensively studied over the past decades due to its significant applications in determining the scattering properties of satellites within the ionosphere [6], calibrating the frequency poles of radar scattering [7], and evaluating the radar cross section (RCS) of absorbing materials [8], among others. Consequently, the accurate and efficient computation of scattered EM fields from the composite spherical structures is of critical importance.

Traditional approaches for computing EM scattering from multilayered spheres predominantly utilize the Mie series method [9], [10], which provides exact analytical solutions for canonical concentric spherical structures subjected to plane wave excitation. However, extending this method to more generalized scenarios, such as excitations by electric or magnetic dipoles, necessitates the application of addition theorems [11]. This leads to expressions involving high-order spherical Bessel and Hankel functions, whose numerical evaluation can become challenging when high accuracy is demanded [12], [13]. Furthermore, when the geometries of the scatterers deviate from strict concentricity or standard 3-D spheres, analytical solutions become intractable [14]. Although, in principle, the analytical formulation remains feasible for composite concentric structures incorporating highly conductive metallic layers within dielectric ones under plane-wave excitation, the associated numerical implementation requires meticulous handling of large-index special functions, complicating its engineering applications [15]. Therefore, more robust methods that can adapt to general situations for EM scattering from multilayered spheres with arbitrary sources are indispensable.

The most straightforward method is employing the surface integral equation (SIE) when dealing with scatterers composed of homogeneous materials. While either a single electric field integral equation (EFIE) or a magnetic field integral equation (MFIE) can independently solve EM scattering problems for homogeneous objects [16], [17], the combined field integral

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equation (CFIE), which results from their superposition, is favored for its enhanced solution stability and reliability [18]. An alternative hybrid method to achieve precise and stable solutions for EM scattering from homogeneous 3-D objects involves the superposition of the external and internal equations of the EFIE and MFIE, respectively, resulting in the well-known Poggio–Miller–Chang–Harrington–Wu–Tsai (PMCHWT) equations [19], [20], [21]. These integral equations (IEs) typically lack analytical solutions and thus are discretized using Rao–Wilton–Glisson (RWG) basis functions, subsequently solved via the method of moments (MoMs). Although enhanced numerical techniques based on MoM, such as the multilevel fast multipole algorithm (MLFMA) [22], can reduce computational time from $O(N^2)$ to $O(N \log N)$, where N represents the number of unknowns in the computational domain, the evaluation of EM scattering from large objects remains expensive.

The primary factor contributing to the high computation cost is the representation of unknown equivalent current densities or total EM fields in the IEs using locally spatial-domain basis functions, such as the RWG functions. As a result, the process of solving the discretized weak-form equations using MoM or enhanced algorithms seeks optimized solutions within a finite-size subspace, which is composed of the discrete nodes or edges of all employed basis functions. Under such a circumstance, a high sampling density (SD) in the discretization is essential to guarantee solution accuracy of the original IEs. Previous studies have demonstrated that, for the discretization using first-order basis functions, the spatial-domain numerical models usually require an SD of at least 10 points per wavelength (PPW) to achieve reliable computation accuracy [23], [24]. On the other hand, excessively increasing SD to enhance the computation accuracy may lead to the dense mesh breakdown [25], [26].

The spectral integral method (SIM) can break this limitation of SD and circumvent the dense mesh breakdown by utilizing spectral-domain basis functions to represent unknown equivalent current densities or total EM fields on smooth boundaries. According to the Nyquist sampling theorem, an SD of 2 PPW is able to guarantee the computational accuracy. The application of the spectral-domain boundary integral to EM scattering by 2-D smooth and closed objects was initially proposed in [27]. It involves transforming the spatial-domain convolutional integral in the traditional 2-D SIE into the multiplication of Fourier spectral coefficients of the unknown total fields and Green's functions. Given that the total fields are inherently smooth and periodic on a closed smooth 2-D boundary, they can be represented by truncated Fourier series. In contrast, the 2-D scalar Green's functions approach infinity when the source point and field point coincide on the boundary, resulting in a singularity that necessitates an infinite number of terms in the Fourier expansion. This issue is resolved by subtracting the singularity from Green's function and then expanding the remaining smooth part using a truncated Fourier series. The details on this singularity extraction technique are available in [27]. Due to its high computational efficiency and exponential convergence accuracy, the SIM has been extensively applied in recent years. Applications mainly include the fast computation of EM scattering from smooth and closed 2-D circular or elliptical cylinders [28], [29], [30], in conjunction with other numerical methods, such as the finite element method (FEM) and spectral element method (SEM) [31], [32], [33], [34], as well as the computation of EM scattering from 3-D bodies of revolution [35], [36]. However, these studies have predominantly focused on SIM applications to smooth 2-D objects with EM fields and Green's functions expanded by 1-D Fourier spectra.

In our previous work [37], we extended the SIM to EM scattering from a fully 3-D sphere for the first time. Both the EFIE for a single metallic sphere and the PMCHWT equations for a single dielectric sphere are transformed into the spectral domain by expanding the surface current densities and dyadic Green's functions (DGFs) using vector spherical harmonics (VSH). The spectral expansions of the DGFs are analytical, with their singularities manifested by an infinite number of the VSH series terms. In this study, building upon the results presented in [37], we develop the multilayer spectral-domain PMCHWT-EFIE framework specifically designed for multilayered concentric spheres with a metallic innermost layer and an arbitrary number of surrounding dielectric layers. Both the EM self-coupling at each layer interface and the EM mutual coupling between two adjacent layer interfaces are directly formulated in the spectral domain. The excitation sources, which can be electric dipoles, magnetic dipoles, or plane waves, may be positioned within any arbitrary layer. Once the spectral-domain impedance matrix is assembled, the VSH spectral coefficients of the surface current densities at all layer boundaries are directly solved, and the scattered fields at the receiver array, which may also be positioned within any arbitrary layer, can be immediately obtained by multiplying the spectral-domain radiation matrix with the coefficient vector. In addition, although the Mie series also can handle EM scattering from multilayered concentric spheres, it is less advantageous than SIM in terms of adaptability, practicality, and computational efficiency. The Mie series is widely employed to compute scattered fields in the far zone when excitation sources are plane waves. However, when these sources are replaced with multiple dipoles with varying polarizations, the computational cost of the Mie series increases significantly. This is due to the necessity of employing the generalized Lorenz–Mie theory, which requires the addition theorem and a complete re-evaluation of the entire series for each source or polarization [38]. Meanwhile, the Mie series often encounters convergence issues when applied to the evaluation of near-zone EM fields or surface current densities on scatterers [39]. By contrast, SIM is more general since it originates from spatial-domain IEs and thus handles the aforementioned problems with greater reliability and efficiency, as will be demonstrated in Section III. Overall, the SIM approach offers greater advantages over the MoM and Mie series in EM scattering scenarios where the scatterers are smooth spheres, possess larger electrical sizes, and are illuminated by multiple near-field dipole sources.

The remainder of this article is organized as follows. In Section II, we derive the multilayered spectral-domain PMCHWT-EFIE formulas, building upon the single-layer

equations presented in [37]. Their discretization is discussed and the algebraic system matrix is constructed, rendering it suitable for computational implementation. In Section III, three numerical examples are provided to validate the computation accuracy and exponential convergence properties of the multilayered SIM by comparing its outcomes with those obtained from Mie series, the traditional spatial-domain MoM, as well as the commercial software COMSOL Multiphysics, which is based on FEM, and Altair Feko, which integrates physical optics (POs) and MLFMA. In particular, the algorithm's reliability and efficiency are assessed for scenarios where transmitters and receivers are positioned within multiple layers. In Section IV, the multilayered SIM is applied to several EM cloak designs to evaluate their invisibility performance. In Section V, summary, conclusion, and future work are presented.

II. Formulation

In this section, we derive the spectral-domain PMCHWT-EFIE for EM scattering from multilayered concentric dielectric spheres surrounding a metal core, based on the formulas presented in [37, Sec. II-B]. The vectorial surface equivalent current densities and DGFs at each spherical boundary are expanded using the VSH series. Additionally, the DGFs that describe the mutual coupling between two adjacent boundaries, which have no singularities, are expanded using the VSH series. Then, the continuous spectral-domain impedance matrix is constructed by incorporating both the diagonal submatrices that formulate the EM self-coupling within each spherical boundary and off-diagonal submatrices that represent the mutual coupling between two adjacent boundaries. The spectral-domain radiation matrix is constructed to facilitate the computation of scattered EM fields collected in an arbitrary layer. Finally, the algebraic impedance and radiation matrices are assembled by discretizing the continuous matrices using perturbed spherical Fibonacci grids (SFGs).

A. Multilayered Spectral-Domain PMCHWT-EFIE

As shown in Fig. 1, it is assumed that there are a total of N concentric spherical boundaries and $N+1$ homogeneous materials. The constitutive parameters of these materials, ranging from the outermost to the penultimate layer, are denoted as $(\epsilon_1,\mu_1),\ldots,(\epsilon_N,\mu_N)$. The innermost sphere is a perfect electric conductor (PEC). Given that the surface equivalent current densities on all boundaries $S_1,\ldots S_N$, as shown in Fig. 1(b), are periodic functions of the solid angle with a cycle of 4π, they can be approximated by the truncated VSH series as follows:

$$\bar{\mathbf{J}}_n(\theta,\phi)=\sum_{l=1}^{L_n}\sum_{m=-l}^{l}a_{l,n}^m\mathbf{m}_l^m(\theta,\phi)+b_{l,n}^m\mathbf{n}_l^m(\theta,\phi) \tag{1a}$$

$$\mathbf{M}_n(\theta,\phi)=\sum_{l=1}^{L_n}\sum_{m=-l}^{l}c_{l,n}^m\mathbf{m}_l^m(\theta,\phi)+d_{l,n}^m\mathbf{n}_l^m(\theta,\phi) \tag{1b}$$

where θ and ϕ are the polar and azimuthal angles, respectively, $\bar{\mathbf{J}}_n=\mathbf{J}_n/\eta_0$ and $\mathbf{M}_n$ denote the surface electric current density scaled by the free-space intrinsic impedance η_0 and the surface magnetic current density, respectively, where the subscript n indicates the spherical boundary index, $\mathbf{m}_l^m$ and $\mathbf{n}_l^m$ are the VSH basis functions, with their expressions detailed in [37, Appendix], $a_{l,n}^m$, $b_{l,n}^m$, $c_{l,n}^m$, and $d_{l,n}^m$ are the corresponding expansion coefficients, and L_n is the upper limit of the order l on the nth spherical surface. Notably, $\mathbf{M}$ is absent on the innermost Nth surface due to the adoption of EFIE for EM scattering from the metal core. To address both the EM self-coupling within each spherical boundary and the mutual coupling between adjacent boundaries, we emulate the formula derivations presented in [37, eqs. (8) and (9)] and redefine the spectral-domain operators as follows:

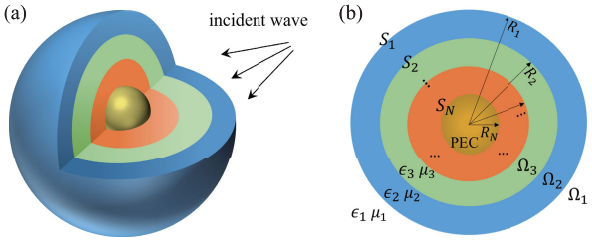


Fig. 1. Multilayered concentric dielectric–metal spheres illuminated by EM waves. (a) 3-D stereogram. (b) 2-D sectional view.

$$\widetilde{\mathcal{L}}_{p,q}^s\{\bar{\mathbf{J}}_q\}=R_q^2\sum_{l=1}^{L_q}\sum_{m=-l}^{l}a_{l,q}^m u_{l,p,q}^{1,s}\mathbf{m}_l^m+b_{l,q}^m u_{l,p,q}^{2,s}\mathbf{n}_l^m+b_{l,q}^m u_{l,p,q}^{3,s}\mathbf{l}_l^m \tag{2a}$$

$$\widetilde{\mathcal{L}}_{p,q}^s\{\mathbf{M}_q\}=R_q^2\sum_{l=1}^{L_q}\sum_{m=-l}^{l}c_{l,q}^m u_{l,p,q}^{1,s}\mathbf{m}_l^m+d_{l,q}^m u_{l,p,q}^{2,s}\mathbf{n}_l^m+d_{l,q}^m u_{l,p,q}^{3,s}\mathbf{l}_l^m \tag{2b}$$

$$\widetilde{\mathcal{K}}_{p,q}^s\{\bar{\mathbf{J}}_q\}=R_q^2\sum_{l=1}^{L_q}\sum_{m=-l}^{l}a_{l,q}^m v_{l,p,q}^{1,s}\mathbf{n}_l^m+a_{l,q}^m v_{l,p,q}^{2,s}\mathbf{l}_l^m+b_{l,q}^m v_{l,p,q}^{3,s}\mathbf{m}_l^m \tag{2c}$$

$$\widetilde{\mathcal{K}}_{p,q}^s\{\mathbf{M}_q\}=R_q^2\sum_{l=1}^{L_q}\sum_{m=-l}^{l}c_{l,q}^m v_{l,p,q}^{1,s}\mathbf{n}_l^m+c_{l,q}^m v_{l,p,q}^{2,s}\mathbf{l}_l^m+d_{l,q}^m v_{l,p,q}^{3,s}\mathbf{m}_l^m \tag{2d}$$

where the subscript $p=1,\ldots,N$ denotes the index of the spherical boundary on which the field point is located, the subscript $q=1,\ldots,N$ represents the index of the spherical boundary on which the equivalent source point is located, and the superscript $s=1,\ldots,N+1$ is the index of the material in which the EM wave propagates and DGFs are evaluated. Because the EM mutual coupling only occurs between two adjacent spherical boundaries, these three indices are constrained by $|p-q|,|s-p|,|s-q|\in\{0,1\}$. Here, R_q is the radius of the qth spherical surface. The coefficients $a_{l,q}^m$, $b_{l,q}^m$, $c_{l,q}^m$, and $d_{l,q}^m$ are the unknowns on the qth spherical surface that are to be determined using the SIM. The coefficients u and v are generated by the VSH expansions of DGFs in the radial direction, as outlined in [37, eqs. (A4) and (A5)]. When $p=q$, they can still be evaluated using [37, eq. (10)] but with the wavenumber k replaced by k_s and the radius R replaced by R_q. However, when $p\neq q$, they must be re-evaluated by

$$u_{l,p,q}^{1,s}=-jk_s h_l^{(2)}(k_s R_p)\,j_l(k_s R_q) \tag{3a}$$

$$u_{l,p,q}^{2,s} = \frac{-j}{k_s R_p R_q} \left[\frac{\partial \left(r h_l^{(2)}(k_s r) \right)}{\partial r} \right]_{r=R_p} \left[\frac{\partial \left(r j_l(k_s r) \right)}{\partial r} \right]_{r=R_q} \tag{3b}$$

$$u_{l,p,q}^{3,s} = -\frac{j\sqrt{l(l+1)}}{k_s R_p R_q} h_l^{(2)}(k_s R_p) \left[\frac{\partial \left(r j_l(k_s r) \right)}{\partial r} \right]_{r=R_q} \tag{3c}$$

$$v_{l,p,q}^{1,s} = \frac{jk_s}{R_p} j_l(k_s r_q) \left[\frac{\partial \left(r h_l^{(2)}(k_s r) \right)}{\partial r} \right]_{r=R_p} \tag{3d}$$

$$v_{l,p,q}^{2,s} = \frac{jk_s\sqrt{l(l+1)}}{R_p} h_l^{(2)}(k_s R_p) j_l(k_s R_q) \tag{3e}$$

$$v_{l,p,q}^{3,s} = \frac{jk_s}{R_q} h_l^{(2)}(k_s R_p) \left[\frac{\partial \left(r j_l(k_s r) \right)}{\partial r} \right]_{r=R_q} \tag{3f}$$

if $p < q$, and

$$u_{l,p,q}^{1,s} = -jk_s h_l^{(2)}(k_s R_q) j_l(k_s R_p) \tag{4a}$$

$$u_{l,p,q}^{2,s} = \frac{-j}{k_s R_p R_q} \left[\frac{\partial \left(r h_l^{(2)}(k_s r) \right)}{\partial r} \right]_{r=R_q} \left[\frac{\partial \left(r j_l(k_s r) \right)}{\partial r} \right]_{r=R_p} \tag{4b}$$

$$u_{l,p,q}^{3,s} = -\frac{j\sqrt{l(l+1)}}{k_s R_p R_q} j_l(k_s R_p) \left[\frac{\partial \left(r h_l^{(2)}(k_s r) \right)}{\partial r} \right]_{r=R_q} \tag{4c}$$

$$v_{l,p,q}^{1,s} = \frac{jk_s}{R_p} h_l^{(2)}(k_s R_q) \left[\frac{\partial \left(r j_l(k_s r) \right)}{\partial r} \right]_{r=R_p} \tag{4d}$$

$$v_{l,p,q}^{2,s} = \frac{jk_s\sqrt{l(l+1)}}{R_p} h_l^{(2)}(k_s R_q) j_l(k_s R_p) \tag{4e}$$

$$v_{l,p,q}^{3,s} = \frac{jk_s}{R_q} j_l(k_s R_p) \left[\frac{\partial \left(r h_l^{(2)}(k_s r) \right)}{\partial r} \right]_{r=R_q} \tag{4f}$$

if $p > q$. Then, following a similar methodology as employed to derive the single-layer spectral-domain EFIE and PMCHWT equations presented in [37, eqs. (11) and (12)], we obtain the multilayered spectral-domain PMCHWT-EFIE as follows:

$$\mathbf{F} = \mathbf{X}\mathbf{C} = \begin{bmatrix} \mathbf{F}_1 \\ \vdots \\ \mathbf{F}_n \\ \vdots \\ \mathbf{F}_N \end{bmatrix} = \begin{bmatrix} \mathbf{X}_{11} & \mathbf{X}_{12} & & & \mathbf{0} \\ \mathbf{X}_{21} & \mathbf{X}_{22} & \mathbf{X}_{23} & & \\ & \mathbf{X}_{32} & \ddots & \ddots & \\ & & \ddots & \ddots & \mathbf{X}_{N-1,N} \\ \mathbf{0} & & & \mathbf{X}_{N,N-1} & \mathbf{X}_{N,N} \end{bmatrix} \begin{bmatrix} \mathbf{C}_1 \\ \vdots \\ \mathbf{C}_n \\ \vdots \\ \mathbf{C}_N \end{bmatrix} \tag{5}$$

where $\mathbf{F}_{1\cdots N-1} \in \mathbb{C}^{4\times 1}$ and $\mathbf{F}_N \in \mathbb{C}^{2\times 1}$ are composed of the spatial-domain incident fields on the nth spherical surface and are constructed as follows:

$$\mathbf{F}_n = \begin{cases} \left[E_{\phi,n}^{inc} \quad -E_{\theta,n}^{inc} \quad \eta_0 H_{\phi,n}^{inc} \quad -\eta_0 H_{\theta,n}^{inc} \right]^T, & \text{if } 1 \le n \le N-1 \\ \left[E_{\phi,n}^{inc} \quad -E_{\theta,n}^{inc} \right]^T, & \text{if } n = N \end{cases} \tag{6}$$

in which $\mathbf{F}_N$ is slightly different since the EFIE only uses the incident electric fields. Similarly, the coefficient vectors $\mathbf{C}_{1\cdots N-1} \in \mathbb{C}^{4\times 1}$ and $\mathbf{C}_N \in \mathbb{C}^{2\times 1}$ are composed of the spectral coefficients of VSH expansions of the surface current densities and are constructed as follows:

$$\mathbf{C}_n = \begin{cases} \left[a_{l,n}^m \quad b_{l,n}^m \quad c_{l,n}^m \quad d_{l,n}^m \right]^T, & \text{if } 1 \le n \le N-1 \\ \left[a_{l,n}^m \quad b_{l,n}^m \right]^T, & \text{if } n = N. \end{cases} \tag{7}$$

The matrix $\mathbf{X}$ is block-tridiagonal. The diagonal submatrix $\mathbf{X}_{nn}$ has the dimensions of 4×4 and is constructed in accordance with [37, eq. (12)] for the range $1 \le n \le N-1$. Conversely, when $n = N$, the submatrix $\mathbf{X}_{nn}$ is 2×2 and is assembled following [37, eq. (11)]. The off-diagonal submatrix $\mathbf{X}_{pq}$, when $p \ne q$, represents the mutual coupling between two adjacent spherical boundaries and exhibits a structure analogous to the matrices described in [37, eqs. (11) and (12)]. However, the superposition of the internal and external equations is unnecessary for a dielectric boundary. Therefore, when $p + 1 = q < N$, based on the formulas given in (3), it can be constructed as follows:

$$\mathbf{X}_{pq} = R_q^2 \sum_{l=1}^{L_q} \sum_{m=-l}^{l} \frac{1}{\sqrt{l(l+1)}} \times \begin{bmatrix} s_l^1 \frac{\partial Y_l^m}{\partial \theta} & -s_l^2 \frac{1}{\sin\theta} \frac{\partial Y_l^m}{\partial \phi} & t_l^1 \frac{1}{\sin\theta} \frac{\partial Y_l^m}{\partial \phi} & -t_l^3 \frac{\partial Y_l^m}{\partial \theta} \\ s_l^1 \frac{1}{\sin\theta} \frac{\partial Y_l^m}{\partial \phi} & s_l^2 \frac{\partial Y_l^m}{\partial \theta} & -t_l^1 \frac{\partial Y_l^m}{\partial \theta} & -t_l^3 \frac{1}{\sin\theta} \frac{\partial Y_l^m}{\partial \phi} \\ -t_l^1 \frac{1}{\sin\theta} \frac{\partial Y_l^m}{\partial \phi} & t_l^3 \frac{\partial Y_l^m}{\partial \theta} & w_l^1 \frac{\partial Y_l^m}{\partial \theta} & -w_l^2 \frac{1}{\sin\theta} \frac{\partial Y_l^m}{\partial \phi} \\ t_l^1 \frac{\partial Y_l^m}{\partial \theta} & t_l^3 \frac{1}{\sin\theta} \frac{\partial Y_l^m}{\partial \phi} & w_l^1 \frac{1}{\sin\theta} \frac{\partial Y_l^m}{\partial \phi} & w_l^2 \frac{\partial Y_l^m}{\partial \theta} \end{bmatrix} \tag{8}$$

in which $s_l^1 = jk_0\mu_s u_{l,p,q}^{1,s}$, $s_l^2 = jk_0\mu_s u_{l,p,q}^{2,s}$, $t_l^1 = v_{l,p,q}^{1,s}$, $t_l^3 = v_{l,p,q}^{3,s}$, $w_l^1 = jk_0\epsilon_s u_{l,p,q}^{1,s}$, and $w_l^2 = jk_0\epsilon_s u_{l,p,q}^{2,s}$. Here, k_0 is the wavenumber in free space. The superscript and subscript s denotes the material index and is equivalent to q. When the condition $p + 1 = q = N$ is met, the magnetic current density $\mathbf{M}$ at the metal surface vanishes. Consequently, $\mathbf{X}_{pq}$ is reduced to a 4×2 matrix, incorporating only the first two columns of the matrix presented in (8). The preceding discussion addresses the construction of the submatrix $\mathbf{X}_{pq}$ when $p < q$. If $q + 1 = p < N$, the expression for $\mathbf{X}_{pq}$ remains consistent with that in (8). However, it is imperative to employ the formulas in (4) rather than those in (3) when evaluating the coefficients s_l^1, s_l^2, t_l^1, t_l^3, w_l^1, and w_l^2. On the other hand, when $q + 1 = p = N$, both $\bar{\mathbf{J}}$ and $\mathbf{M}$ on the $(N-1)$th spherical surface radiate electric fields toward the Nth metal surface. Therefore, $\mathbf{X}_{pq}$ is represented as a 2×4 matrix, incorporating only the first two rows of the matrix depicted in (8), with the coefficients s_l^1, s_l^2, t_l^1, and t_l^3 determined using (4) instead of (3). Upon resolving the coefficient vector $\mathbf{C}$ from (5), the scattered fields at the receiver array can be calculated by employing the operators $\widetilde{\mathcal{L}}$ and $\widetilde{\mathcal{K}}$, as defined in (2). When a receiver is positioned between two spherical boundaries, the contributions to the scattered fields from both boundaries are superimposed. Meanwhile, R_p in (3) and (4) must be substituted with R_r, and the variables (θ, ϕ) in the VSH basis functions $\mathbf{l}_l^m$, $\mathbf{m}_l^m$, and $\mathbf{n}_l^m$ must be replaced with (θ_r, ϕ_r). Here, (R_r, θ_r, ϕ_r) denotes the spherical coordinates of the rth receiver.

B. Discretization of the Multilayered Spectral-Domain PMCHWT-EFIE

Equation (5) exhibits continuity. Both the incident fields within the vector $\mathbf{F}$ and the spherical scalar basis function Y_l^m within the matrix $\mathbf{X}$ demonstrate continuous variation with respect to the field point (θ, ϕ). Nevertheless, the total number of unknown VSH spectral coefficients amounts to $4\Sigma_{n=1}^{N-1}\{(L_n+1)^2-1\}+2\{(L_N+1)^2-1\}$ when the surface current densities on the nth spherical boundary are expanded using the VSH basis functions $\mathbf{m}_l^m$ and $\mathbf{n}_l^m$ with an order of L_n. To resolve all the unknown coefficients, it is necessary to discretize (5) and generate an equivalent number of independent equations as the total unknown spectral coefficients. In this work, we also employ the perturbed SFGs, as proposed in [37, Sec. II-C], to independently sample the incident fields on each spherical surface. The truncation threshold L_n of the VSH series for each spherical surface is determined by a prescribed computation accuracy.

C. Assembly of the Algebraic System Matrix for the Multilayered Spectral-Domain PMCHWT-EFIE

We first assume that the surface current densities on the qth spherical boundary are represented through the VSH series with an order of L_q to ensure computational accuracy. Consequently, there are a total of $4M_q = 4(L_q+1)^2-4$ unknowns for the PMCHWT equations on the qth surface if $1 \le q \le N-1$ and $2M_N = 2(L_N+1)^2-2$ unknowns for the EFIE on the Nth metal surface. To produce an equivalent number of equations from (5), it is necessary to evaluate the subvector $\mathbf{F}_p$ and the submatrix $\mathbf{X}_{pq}$ at an identical number of SFGs on each spherical surface. For the pth surface, it is assumed that these SFGs are located at $(\theta_{i,p}, \phi_{i,p})$ with $i = 1, \ldots, 4M_p$ for $1 \le p \le N-1$ and $i = 1, \ldots, 2M_p$ for $p = N$. Through sampling at these grids, (5) can be discretized and compactly written as follows:

$$\mathbf{V} = \mathbf{ZI} \tag{9}$$

in which $\mathbf{V}$ corresponds to $\mathbf{F}$, as defined in (5), and is structured as a cascade of N column subvectors, denoted as $\mathbf{V}_1, \ldots, \mathbf{V}_N$. Similarly, $\mathbf{Z}$ corresponds to $\mathbf{X}$ and is composed of $3N-2$ submatrices, specifically $\mathbf{Z}_{11}, \mathbf{Z}_{12}, \mathbf{Z}_{21}, \ldots, \mathbf{Z}_{N,N-1}, \mathbf{Z}_{N,N}$. Furthermore, $\mathbf{I}$ corresponds to $\mathbf{C}$ and is also arranged as a cascade of N column subvectors, namely, $\mathbf{I}_1, \ldots, \mathbf{I}_N$. Given that the incident fields at each SFG encompass various components, $\mathbf{V}_p$ can be further decomposed into

$$\mathbf{V}_p = \left[\mathbf{V}_p^1 \cdots \mathbf{V}_p^i \cdots \mathbf{V}_p^{M_p}\right]^T \tag{10}$$

in which

$$\mathbf{V}_p^i = \begin{cases} \left[E_{\phi,p}^{inc}(\theta_{i,p},\phi_{i,p}) \quad -E_{\theta,p}^{inc}(\theta_{i,p},\phi_{i,p}) \quad \eta_0 H_{\phi,p}^{inc}(\theta_{i,p},\phi_{i,p}) \right. \\ \left. -\eta_0 H_{\theta,p}^{inc}(\theta_{i,p},\phi_{i,p})\right], & \text{if } 1 \le p \le N-1 \\ \left[E_{\phi,p}^{inc}\left(\theta_{i,p},\phi_{i,p}\right) \quad -E_{\theta,p}^{inc}\left(\theta_{i,p},\phi_{i,p}\right)\right], & \text{if } p = N \end{cases} \tag{11}$$

where p denotes the index of the spherical surface on which the discrete field points are located, and i represents the index of the field point on the pth surface. The submatrix $\mathbf{Z}_{pq}$ can be constructed in a manner analogous to that described in [37, Sec. II-D] when $p = q$. For the off-diagonal submatrix where $p \ne q$, it can be further decomposed into

$$\mathbf{Z}_{pq} = \begin{cases} \left[\mathbf{Z}_{a,pq} \quad \mathbf{Z}_{b,pq}\right], & \text{if } p+1 = q = N \\ \left[\mathbf{Z}_{a,pq} \quad \mathbf{Z}_{b,pq} \quad \mathbf{Z}_{c,pq} \quad \mathbf{Z}_{d,pq}\right], & \text{otherwise} \end{cases} \tag{12}$$

where $\mathbf{Z}_{a,b,c,d,pq}$ has the dimensions of $2M_p \times M_q$ if $p = q+1 = N$. In all other cases, its dimensions are $4M_p \times M_q$. It is assembled as follows:

$$\mathbf{Z}_{a,b,c,d,pq} = \begin{bmatrix} \mathbf{Z}_{a,b,c,d,pq}^{1,1} & \cdots & \mathbf{Z}_{a,b,c,d,pq}^{1,l} & \cdots & \mathbf{Z}_{a,b,c,d,pq}^{1,L_q} \\ \vdots & \ddots & & & \vdots \\ \mathbf{Z}_{a,b,c,d,pq}^{i,1} & & \mathbf{Z}_{a,b,c,d,pq}^{i,l} & & \mathbf{Z}_{a,b,c,d,pq}^{i,L_q} \\ \vdots & & & \ddots & \vdots \\ \mathbf{Z}_{a,b,c,d,pq}^{M_p,1} & \cdots & \mathbf{Z}_{a,b,c,d,pq}^{M_p,l} & \cdots & \mathbf{Z}_{a,b,c,d,pq}^{M_p,L_q} \end{bmatrix} \tag{13}$$

in which

$$\mathbf{Z}_{a,b,c,d,pq}^{i,l} = \left[\mathbf{Z}_{a,b,c,d,pq}^{i,-l} \cdots \mathbf{Z}_{a,b,c,d,pq}^{i,m} \cdots \mathbf{Z}_{a,b,c,d,pq}^{i,l}\right] \tag{14}$$

where $\mathbf{Z}_{a,b,c,d,pq}^{i,m}$ has the dimensions of 2×1 if $p = q+1 = N$. Otherwise, its dimensions are 4×1. It is constructed by extracting specific elements from the 4×4 matrix $\mathbf{X}_{pq}$ in (8), evaluated at the ith sampling points on the pth spherical surface. The four submatrices are assembled as follows:

$$\mathbf{Z}_{a,pq}^{i,m} = \begin{bmatrix} X_{pq}(1,1) \\ X_{pq}(2,1) \end{bmatrix}_{\substack{\theta=\theta_{i,p} \\ \phi=\phi_{i,p}}} \mathbf{Z}_{b,pq}^{i,m} = \begin{bmatrix} X_{pq}(1,2) \\ X_{pq}(2,2) \end{bmatrix}_{\substack{\theta=\theta_{i,p} \\ \phi=\phi_{i,p}}} \tag{15a}$$

$$\mathbf{Z}_{c,pq}^{i,m} = \begin{bmatrix} X_{pq}(1,3) \\ X_{pq}(2,3) \end{bmatrix}_{\substack{\theta=\theta_{i,p} \\ \phi=\phi_{i,p}}} \mathbf{Z}_{d,pq}^{i,m} = \begin{bmatrix} X_{pq}(1,4) \\ X_{pq}(2,4) \end{bmatrix}_{\substack{\theta=\theta_{i,p} \\ \phi=\phi_{i,p}}} \tag{15b}$$

if $p = q+1 = N$. Otherwise, they are assembled as follows:

$$\mathbf{Z}_{a,pq}^{i,m} = \begin{bmatrix} X_{pq}(1,1) \\ X_{pq}(2,1) \\ X_{pq}(3,1) \\ X_{pq}(4,1) \end{bmatrix}_{\substack{\theta=\theta_{i,p} \\ \phi=\phi_{i,p}}} \mathbf{Z}_{b,pq}^{i,m} = \begin{bmatrix} X_{pq}(1,2) \\ X_{pq}(2,2) \\ X_{pq}(3,2) \\ X_{pq}(4,2) \end{bmatrix}_{\substack{\theta=\theta_{i,p} \\ \phi=\phi_{i,p}}} \tag{16a}$$

$$\mathbf{Z}_{c,pq}^{i,m} = \begin{bmatrix} X_{pq}(1,3) \\ X_{pq}(2,3) \\ X_{pq}(3,3) \\ X_{pq}(4,3) \end{bmatrix}_{\substack{\theta=\theta_{i,p} \\ \phi=\phi_{i,p}}} \mathbf{Z}_{d,pq}^{i,m} = \begin{bmatrix} X_{pq}(1,4) \\ X_{pq}(2,4) \\ X_{pq}(3,4) \\ X_{pq}(4,4) \end{bmatrix}_{\substack{\theta=\theta_{i,p} \\ \phi=\phi_{i,p}}} \tag{16b}$$

where $\theta_{i,p}$ and $\phi_{i,p}$ are the polar and azimuthal coordinates, respectively, of the ith SFG on the pth spherical surface. The two numbers inside the bracket are row and column indices, respectively, of the matrix $\mathbf{X}_{pq}$. The column vector $\mathbf{I}$ in (9) can be constructed by concatenating the spectral coefficient vectors across all spherical boundaries

$$\mathbf{I} = \left[\mathbf{I}_1 \cdots \mathbf{I}_q \cdots \mathbf{I}_N\right]^T \tag{17}$$

in which

$$\mathbf{I}_q = \begin{cases} \left[\mathbf{a}_q^1 \cdots \mathbf{a}_q^l \cdots \mathbf{a}_q^{L_q} \ \mathbf{b}_q^1 \cdots \mathbf{b}_q^l \cdots \mathbf{b}_q^{L_q} \ \mathbf{c}_q^1 \cdots \mathbf{c}_q^l \cdots \right. \\ \left. \quad \mathbf{c}_q^{L_q} \ \mathbf{d}_q^1 \cdots \mathbf{d}_q^l \cdots \mathbf{d}_q^{L_q}\right], & \text{if } 1 \le q \le N-1 \\ \left[\mathbf{a}_q^1 \cdots \mathbf{a}_q^l \cdots \mathbf{a}_q^{L_q} \ \mathbf{b}_q^1 \cdots \mathbf{b}_q^l \cdots \mathbf{b}_q^{L_q}\right], & \text{if } q = N \end{cases} \tag{18}$$

where

$$\begin{cases} \mathbf{a}_q^l = \left[a_{l,q}^{-l} \cdots a_{l,q}^{m} \cdots a_{l,q}^{l}\right] \in \mathbb{C}^{1\times(2l+1)} \\ \mathbf{b}_q^l = \left[b_{l,q}^{-l} \cdots b_{l,q}^{m} \cdots b_{l,q}^{l}\right] \in \mathbb{C}^{1\times(2l+1)} \\ \mathbf{c}_q^l = \left[c_{l,q}^{-l} \cdots c_{l,q}^{m} \cdots c_{l,q}^{l}\right] \in \mathbb{C}^{1\times(2l+1)} \\ \mathbf{d}_q^l = \left[d_{l,q}^{-l} \cdots d_{l,q}^{m} \cdots d_{l,q}^{l}\right] \in \mathbb{C}^{1\times(2l+1)} \end{cases} \tag{19}$$

in which the superscript l takes values within the range $[1, L_q]$. Upon determining $\mathbf{I}$ from (9), the scattered fields at the receiver array can be calculated by multiplying $\mathbf{I}$ with the spectral-domain radiation matrix, which is structurally analogous to that presented in [37, eq. (25)]. However, when a receiver is positioned between two adjacent spherical surfaces, the current densities on both adjacent surfaces contribute simultaneously to the scattered fields, necessitating the superimposition of two matrices similar to that in [37, eq. (25)]. Given the straightforward nature of this procedure, it will not be elaborated upon here.

III. Numerical Validation

In this section, we present three numerical examples to validate the computational accuracy of the proposed multilayered spectral-domain PMCHWT-EFIE (hereinafter referred to as multilayered SIM) and demonstrate its effectiveness in analyzing EM scattering from dielectric–metal concentric spherical structures. Three cases are examined, each with distinct geometric dimensions and excitation configurations. In the first case, a plane wave with an operating frequency of 1 GHz is incident upon a coated metal sphere characterized by varying electrical sizes and constitutive parameters. The calculated RCS values are checked against those obtained through the Mie series evaluation to verify the correctness of SIM. In the second case, both the single electric dipole transmitter and the receiver array are placed in the outermost layer. The computational results are verified against those obtained from the traditional spatial-domain MoM and COMSOL. The third case involves scatterers with considerably larger electrical sizes, employs multiple excitation sources and receivers distributed across all layers, and is compared with the results obtained using Feko. In these last two cases, the operating frequency is 300 MHz, and the superior efficiency of SIM compared to MoM is confirmed. The Mie series is hardly applicable in this context due to the utilization of multiple dipole sources and the evaluation of surface current densities. All simulations and numerical computations are conducted on a workstation equipped with a 48-core Intel Xeon 6248R 3.0-GHz CPU and 1024 GB of memory. The SIM and MoM codes used for the comparisons are executed in a single-threaded mode. However, the commercial software packages COMSOL and Feko are executed in parallel using all CPU cores.

A. EM Scattering From a Coated Metal Sphere Under Plane Wave Excitation

As illustrated in Fig. 2, the concentric scatterers comprise a spherical metal core and a thin coating layer with the complex

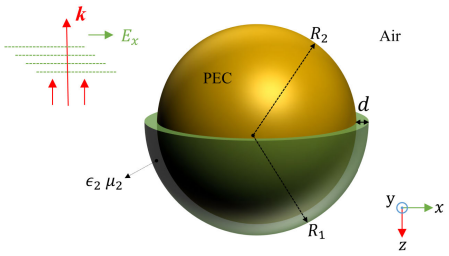


Fig. 2. Coated metal sphere with its center located at the origin is illuminated by plane waves.

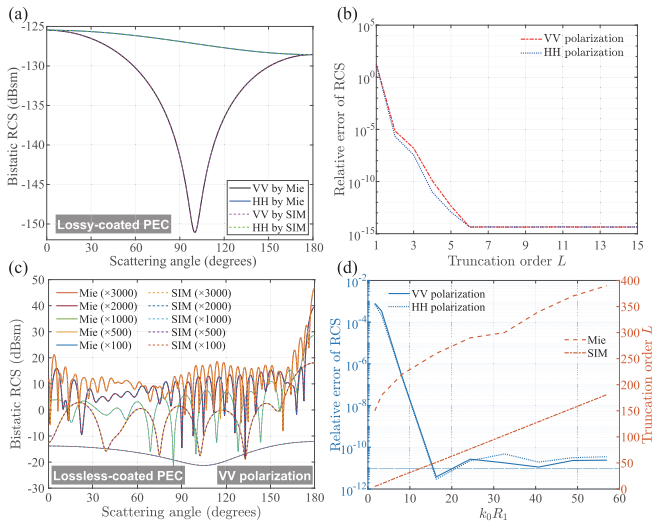


Fig. 3. Comparisons of RCS values computed by SIM and Mie series for EM scattering from coated metal spheres with different electrical sizes. (a) RCS values for spheres with extremely small electrical sizes. (b) Variations of SIM errors with the increment of the truncation order L. (c) RCS values for spheres with different electrical sizes. (d) Variations of SIM errors with the increment of radii of spheres when a fixed SD = 4 PPW is adopted. Red lines denote the corresponding truncation order L used in computation.

relative permittivity of $\epsilon_2 = 2.0 - j1.5$ and the permeability $\mu_2 = 1.0$. The thickness of this layer is $d = 0.001\lambda_0$. The radius of the inner metal sphere is $R_2 = 0.01\lambda_0/2\pi$, while the radius of the outer sphere is $R_1 = R_2 + d$. These parameters indicate that the scattering model possesses both a very small electrical size and a thin layer. The Mie-series results serve as benchmarks, with their truncation criterion being uniquely determined by the maximum order L that the computer can achieve to prevent significant numerical underflow or overflow.

As shown in Fig. 3(a), the RCS values obtained via SIM for both polarizations exhibit strong agreement with those derived from the Mie series. Fig. 3(b) depicts the variation of relative errors with the truncation order L of SIM. It is evident that the relative RCS error approaches 10^{-15} with merely $L = 6$ expansion terms in the SIM. These findings demonstrate the high efficiency of the proposed SIM and its rapid convergence in computing EM scattering from smooth 3-D concentric spheres characterized by extremely small electrical sizes and thin coating layers.

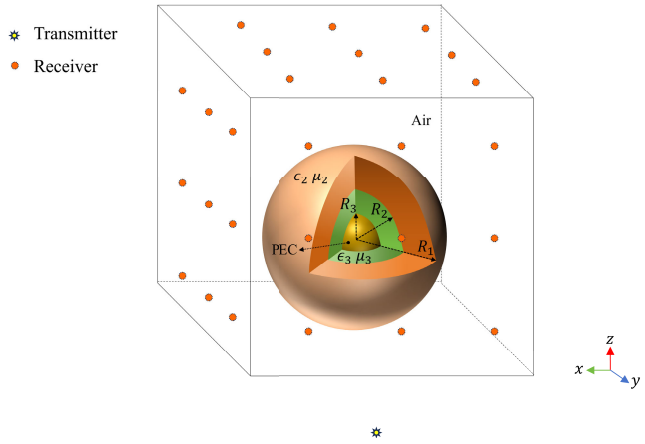


Fig. 4. Three-layer dielectric–metal spheres with their centers located at the origin illuminated by EM waves emitted from a 3-D infinitesimal electric dipole transmitter. The receivers are located on six faces of a virtual cube.

To further investigate the impact of dimension contrast between consecutive layers on the numerical stability of SIM, R_1 and R_2 are incrementally increased using a consistent scaling factor, thereby generating a series of electric sizes ranging from small to large. Meanwhile, to enhance the material contrast between the metal core and the coating layer, the conductivity of the coating layer is eliminated, and its relative permittivity is set to $\epsilon_r = 2.0$. Fig. 3(c) shows the RCS values computed by SIM, which are compared with the results from the Mie series for various typical enlarging factors of the radii, such as 100, 500, 1000, 2000, and 3000. It is evident that the SIM results agree well with the Mie-series results, regardless of the magnitude of the difference $(R_1 - R_2)$. Fig. 3(d) illustrates the variation in RCS errors with the radius R_1 when SD in the discretization is maintained at 4 PPW. The corresponding truncation order L values used in the computation of SIM and Mie series are indicated by red lines. It is apparent that the computation errors of SIM are independent of the dimension contrast, provided that the truncation order L is sufficiently large. Notably, the larger errors in SIM results for smaller spheres are attributed to the smaller truncation order L values used in the SIM computation for smaller spheres at a fixed SD value. By increasing the truncation order L of SIM, the computation error can decrease exponentially, as demonstrated in Fig. 3(b). These findings suggest that SIM can maintain numerical stability across a wide range of layer thicknesses without exhibiting any signs of instability or convergence degradation.

B. EM Scattering From Multilayered Spheres Under Single Transmitter Excitation

As shown in Fig. 4, the concentric scatterers consist of three spheres. The outermost material is air. Other spheres have the constitutive parameters $\varepsilon_2 = 4.0$, $\mu_2 = 2.0$, $\sigma_2 = 2.0$ mS/m, $\varepsilon_3 = 3.0$, $\mu_3 = 4.0$, and $\sigma_3 = 4.0$ mS/m. The innermost sphere is metal. The radii of these spheres are $R_1 = 0.3\lambda_0$, $R_2 = 0.2\lambda_0$, and $R_3 = 0.1\lambda_0$, where λ_0 represents the free-space wavelength at the operating frequency. An infinitesimal electric dipole, polarized in the direction of $(1, 1, 1)$, is positioned at $(0, 0, -1.5)$ m. The receivers are placed on the surface of a cube centered at the origin, with dimensions of $1.0 \times 1.0 \times 1.0$ m, enclosing the spherical scatterers. Each face of the cube contains nine uniformly distributed receivers. For instance, the first receiver of the 3×3 array located on the top face at $z = 0.5$ m has the coordinate of $(-0.4, -0.4, 0.5)$ m. The increment intervals between adjacent receivers in both the $\hat{x}$-direction and $\hat{y}$-direction are 0.35 m.

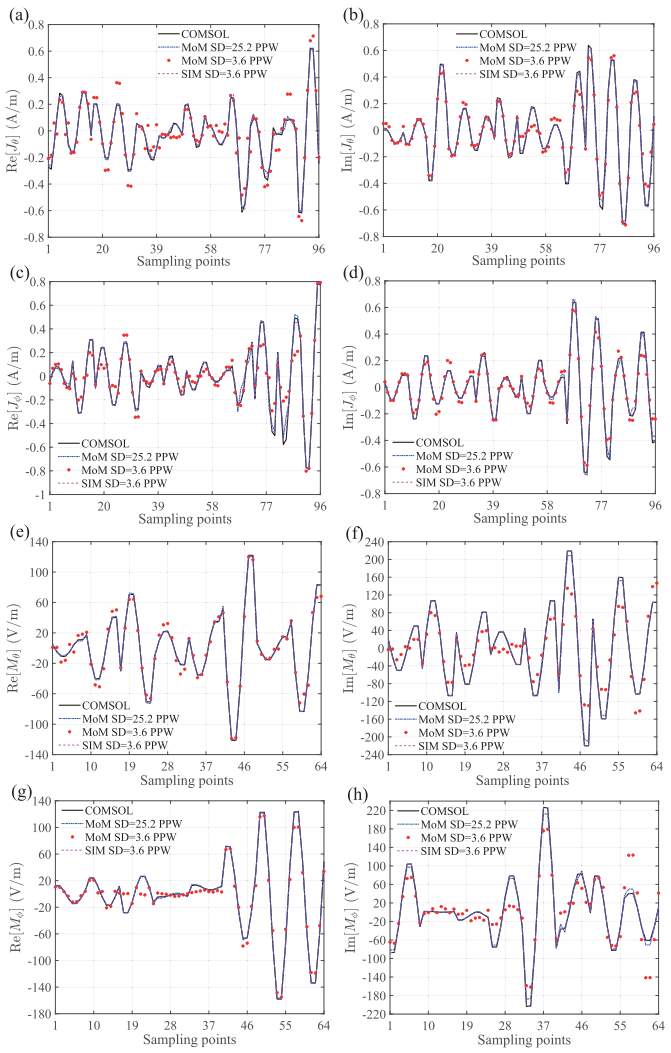


Fig. 5. Comparisons of the equivalent surface electric and magnetic currents computed by COMSOL, MoM, and SIM when the spatial sampling densities take different values. (a) Real part of J_θ. (b) Imaginary part of J_θ. (c) Real part of J_ϕ. (d) Imaginary part of J_ϕ. (e) Real part of M_θ. (f) Imaginary part of M_θ. (g) Real part of M_ϕ. (h) Imaginary part of M_ϕ.

First, let us verify the correctness of the equivalent surface current densities solved from (9). Here, we pick 32 representative sampling points on each spherical surface, arranged in a 4×8 grid with equal intervals of $\pi/4$ in both the $\hat{\theta}$- and $\hat{\phi}$-directions. The spherical angular coordinate of the first sampling point is $(\pi/8, \pi/8)$. The surface current densities at these points, calculated using COMSOL, MoM, and multilayered SIM with different SD values, are shown in

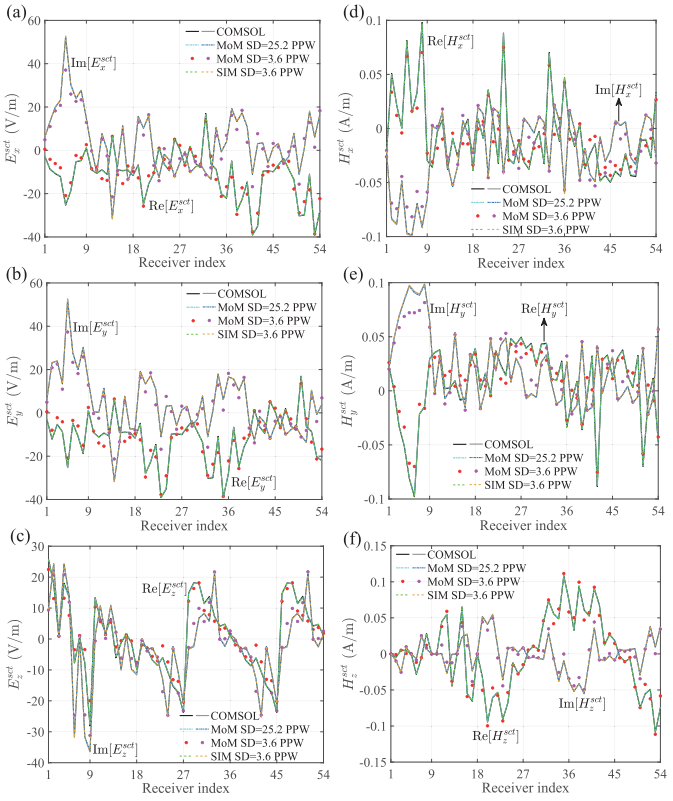


Fig. 6. Comparisons of the scattered EM fields at the receiver array computed by COMSOL, MoM, and SIM when the spatial sampling densities on the sphere surfaces take different values. (a) Real and imaginary parts of E_x^{sct}. (b) Real and imaginary parts of E_y^{sct}. (c) Real and imaginary parts of E_z^{sct}. (d) Real and imaginary parts of H_x^{sct}. (e) Real and imaginary parts of H_y^{sct}. (f) Real and imaginary parts of H_z^{sct}.

Fig. 5. Note the connecting lines between two sampling points should not be interpreted as interpolation or approximation at locations that have not been sampled. Instead, they are used solely to differentiate various data types in figures. The COMSOL results, obtained using highly dense tetrahedral meshes with a total of 29 104 064 degrees of freedom, serve as benchmarks. The MoM, adapted for EM scattering from multilayered nested spheres, is directly modified from the single sphere approach used in [37]. As can be seen, the electric and magnetic current density variations on the sphere surfaces computed by multilayered SIM match those by COMSOL well, even when the SD of SIM is lowered to 3.6 PPW. In contrast, the traditional MoM requires an increase in SD to 25.2 PPW to achieve comparable accuracy. Precise calculations reveal that the relative errors of J_θ for MoM with SD = 25.2 PPW, MoM with SD = 3.6 PPW, and SIM with SD = 3.6 PPW, compared to COMSOL results, are 8.1%, 32.4%, and 4.7%, respectively. The relative errors of J_ϕ for these scenarios are 10.6%, 33.8%, and 5.7%, respectively. Similarly, the relative errors of M_θ are 4.6%, 39.4%, and 0.6%, respectively, while the relative errors of M_ϕ are 6.5%, 37.5%, and 0.9%, respectively. These error metrics indicate that SIM requires an SD of approximately 4 PPW to achieve reliable computational accuracy, whereas MoM necessitates an increase in SD to approximately 25 PPW to maintain computational precision.

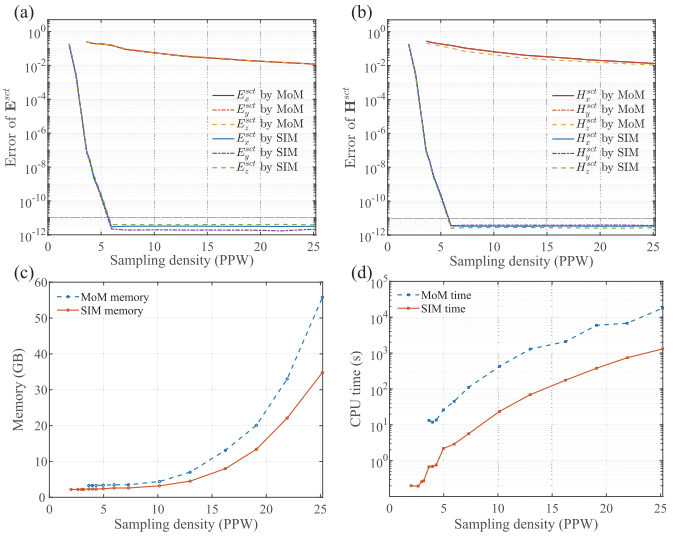


Fig. 7. Comparisons of the computation errors of scattered fields, the memory, and CPU time costs of MoM and SIM for different spatial sampling densities. (a) Errors of scattered electric fields computed by MoM and SIM. (b) Errors of scattered magnetic fields computed by MoM and SIM. (c) Memory consumed by MoM and SIM. (d) CPU time consumed by MoM and SIM.

Then, we proceed to verify the accuracy of the scattered fields at the receiver array as computed by the SIM through comparing these results with those obtained using COMSOL and MoM. Fig. 6 shows the comparisons. Using the COMSOL results as benchmarks, the relative errors for E_x^{sct} with MoM at SD = 25.2 PPW, MoM at SD = 3.6 PPW, and SIM at SD = 3.6 PPW are 1.2%, 24.4%, and 0.5%, respectively. The relative errors for E_y^{sct} are 1.4%, 24.6%, and 0.6%, respectively. For E_z^{sct}, the relative errors are 1.6%, 25.0%, and 0.8%, respectively. The relative errors for H_x^{sct} are 1.6%, 27.9%, and 0.7%, respectively, while for H_y^{sct}, they are 1.5%, 27.8%, and 0.6%, respectively. Finally, the relative errors for H_z^{sct} are 1.2%, 21.1%, and 0.5%, respectively. It is evident that the scattered fields at the receiver array computed by SIM achieve an error magnitude on the order of one-thousandth when the SD is reduced to 3.6 PPW. However, the MoM requires an increased SD of 25.2 PPW to attain a comparable error level. Furthermore, the SIM requires only 0.7 s and 2.3 GB of memory to compute the scattered fields at SD = 3.6 PPW. In contrast, the MoM requires 17 883.2 s and 55.8 GB of memory at SD = 25.2 PPW. Clearly, the SIM demonstrates superior computational efficiency compared to MoM while maintaining similar computational accuracy.

Finally, we assess the exponential convergence accuracy of the multilayered SIM as the spatial SD on the sphere surfaces is incrementally increased. The SIM results at SD = 30.1 PPW serve as benchmarks. Fig. 7(a) and (b) show the variations of relative errors of the scattered fields at the receiver array when SD increases from 2.0 to 25.2 PPW. It is observed that the errors of SIM results decrease exponentially when SD is less than 6.0 PPW, after which they stabilize at the order of 10^{-12}. In contrast, the errors of the MoM results diminish at a slower rate, reaching approximately 1.0% when SD is increased to 25.2 PPW. Fig. 7(c) and (d) show the comparisons of memory

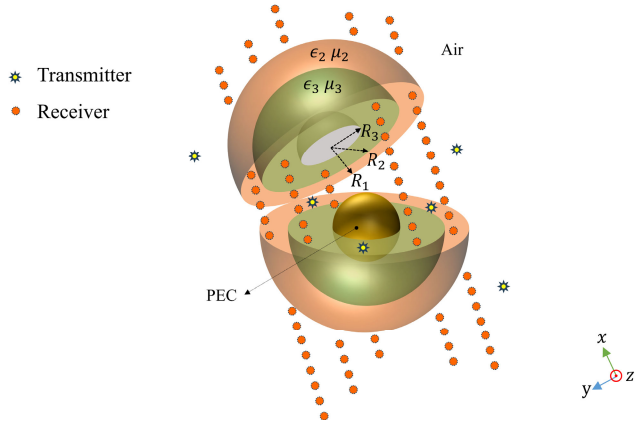


Fig. 8. Three-layer dielectric–metal spheres with their centers located at the origin illuminated by EM waves emitted from six infinitesimal 3-D electric dipole transmitters located in different regions. The receiver array is composed of five straight lines traversing different dielectric layers.

and CPU time costs for SIM and MoM across different sampling densities. Evidently, the discrepancies in both memory and CPU time costs progressively increase with higher sampling densities. A plausible explanation for this phenomenon is that MoM must handle the singularities of DGFs when two discretized grids are adjacent or overlap. Conversely, in the multilayered SIM, the singularities are manifested by the VSH series terms with higher orders, which are truncated by the limited orders of the VSH series of the smooth surface current densities.

C. EM Scattering From Multilayered Spheres Under Multiple Transmitter Excitations

In this case, we employ the multilayered SIM to analyze EM scattering from much larger nested scatterers. As shown in Fig. 8, the three spheres have the radii of $R_1 = 1.5\lambda_0$, $R_2 = 0.8\lambda_0$, and $R_3 = 0.1\lambda_0$. The constitutive parameters of all layers are consistent with those utilized in Case B. Six electric dipole transmitters, each with distinct polarizations, are positioned within different layers. In the outermost air layer, three transmitters are located at coordinates (0, 0, −2) m, (−0.87, −1.5, 1) m, and (0.87, 1.5, 1) m, with corresponding polarizations of (1, 1, 1), (1, 0, 0), and (0, 1, 0), respectively. Two additional transmitters are located within the outer spherical dielectric crown at coordinates (−0.5, −0.86, −0.57) m and (0.5, 0.86, −0.57) m, with polarizations of (1, 0, 1) and (0, 1, 1), respectively. The final transmitter is positioned within the inner spherical dielectric crown at the coordinate (0, 0, 0.45) m, with a polarization of (1, 1, 1). The receiver array consists of five straight lines, each comprising 11 uniformly distributed observation points with an interval of 0.1 m. Three of these receiver-array lines traverse both the outer and inner spherical dielectric crowns. The first receivers of these three lines are located at (2.0, 0.1, −0.5) m, (2.0, 0.5, 0.2) m, and (2.0, −0.5, 0.2) m, while the last receivers are positioned at (−2.0, 0.1, −0.5) m, (−2.0, 0.5, 0.2) m, and (−2.0, −0.5, 0.2) m, respectively. The remaining two lines penetrate solely the outer spherical dielectric crown. The first receivers of these two lines are located at (2.0, −1.1, 0.5) m and (2.0, 1.1, 0.5) m, and the last receivers are situated at (−2.0, −1.1, 0.5) m and (−2.0, 1.1, 0.5) m, respectively. This configuration guarantees the evaluation of the proposed multilayered SIM's adaptability to scenarios where transmitters and receivers are placed in distinct layers.

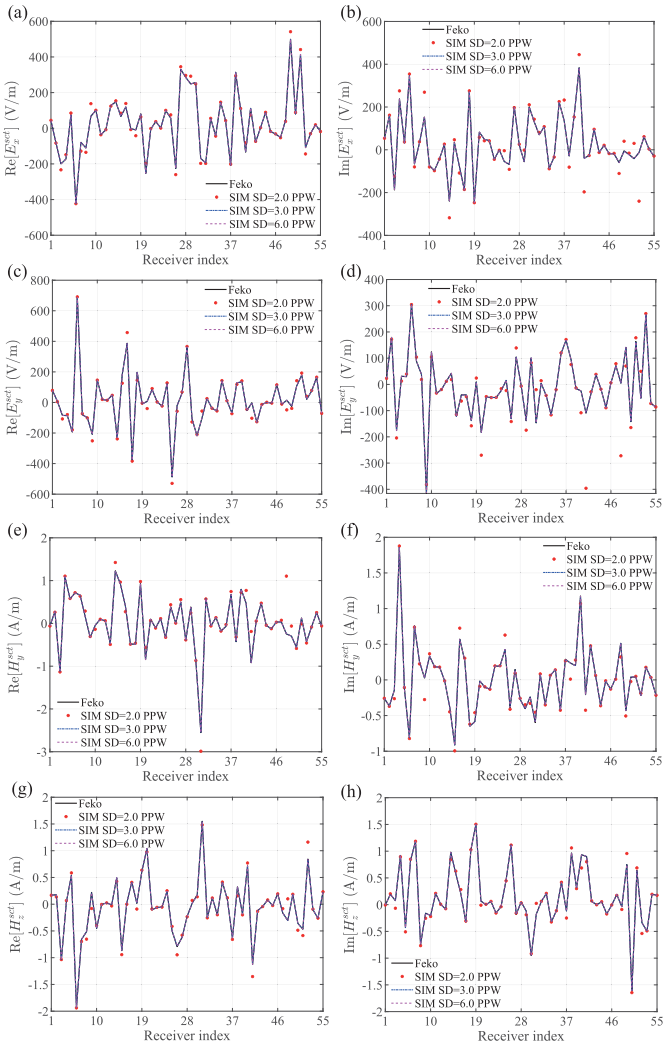


Fig. 9. Comparisons of the scattered electric and magnetic fields at the receiver array computed by Feko and SIM when the spatial sampling densities on the sphere surfaces take different values. (a) Real part of E_x^{sct}. (b) Imaginary part of E_x^{sct}. (c) Real part of E_y^{sct}. (d) Imaginary part of E_y^{sct}. (e) Real part of H_y^{sct}. (f) Imaginary part of H_y^{sct}. (g) Real part of H_z^{sct}. (h) Imaginary part of H_z^{sct}.

Fig. 9 shows the comparisons of the scattered fields at the receiver array, as computed by the multilayered SIM and simulated by Feko. Due to the space limitation, only partial components of the scattered fields are showcased. Feko discretizes the metal core surface using 3016 triangular elements and the two dielectric surfaces using 76 592 triangular elements. This fine discretization guarantees the reliability of the Feko results, which serve as benchmarks in this case. The SIM results exhibit notable discrepancies from the Feko results for SD = 2.0 PPW. However, upon increasing the SD to 3.0 PPW, the SIM and Feko results demonstrate substantial agreement. Further increasing the SD to 6.0 PPW does not result in observable changes to the SIM results. This is corroborated by the precise evaluation of the relative errors of the SIM results in comparison to the Feko results, as detailed

TABLE I
COMPARISONS OF COMPUTATION COSTS BETWEEN FEKO AND MULTILAYERED SIM AS WELL AS THE RELATIVE ERRORS

Method	SD (PPW)	Memory (GB)	Time (s)	Error of $\mathbf{E}^{sct}$			Error of $\mathbf{H}^{sct}$		
				x	y	z	x	y	z
Feko	16.2	432.0	19560.0	–	–	–	–	–	–
SIM	2.0	3.6	16.5	25.68%	32.18%	12.26%	28.62%	32.38%	15.87%
	3.0	8.0	92.2	0.53%	0.52%	0.44%	0.46%	0.52%	0.57%
	6.0	85.3	1900.8	0.52%	0.50%	0.43%	0.46%	0.50%	0.55%

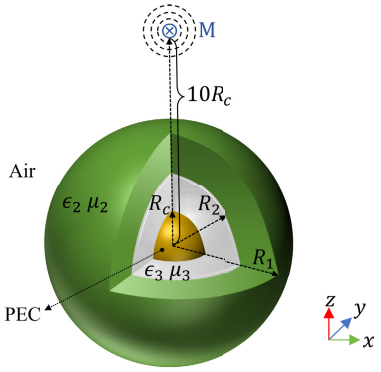


Fig. 10. Three-layer cloaked sphere with a metal core and two dielectric shells illuminated by EM waves emitted from a magnetic dipole.

in Table I. The errors decrease significantly when the SD is increased from 2.0 to 3.0 PPW, but remain nearly constant when the SD is further increased from 3.0 to 6.0 PPW. On the other hand, the SIM significantly outperforms Feko in terms of computation time and memory consumption. This is primarily because SIM is implemented in the spectral domain with low sampling densities, whereas Feko, after all, employs the spatial-domain MoM, which incurs high computational costs for scatterers with large electrical sizes, even when utilizing the MLFMA and PO. These numerical experiments testify that the proposed multilayered SIM is capable of reliably and efficiently addressing EM scattering from nested spherical scatterers with transmitters and receivers placed in arbitrary layers.

IV. APPLICATIONS OF MULTILAYERED SIM

In this section, we demonstrate the practical applicability of the proposed multilayered SIM through two cases of EM cloaking designs for concentric dielectric–metal spherical structures operating at distinct frequencies. The first case involves a metal sphere cloaked by two dielectric shells under magnetic dipole excitation. The second case examines a metal sphere cloaked by multiple dielectric layers under plane-wave illumination. In both cases, the cloaking performance is assessed by analyzing the field distribution in the vicinity of the scatterer and the reduction in scattering cross section (SCS) or RCS. The SCS is related to RCS through $\sigma_{SCS} = 10\log_{10}[\sigma_{RCS}/(\pi R_c^2)]$, where R_c is the radius of the core sphere and πR_c^2 is the geometrical projection area.

A. Metal Sphere Cloaked by Two Dielectric Shells Under Magnetic Dipole Excitation

As shown in Fig. 10, we first consider an EM cloaking structure comprising a metal core surrounded by two concentric dielectric shells, which are illuminated by waves emitted from a magnetic dipole. The operating frequency is 1.5 GHz. The geometric and constitutive parameters for the typical designs A and B are adapted from [40] and are summarized in Table II. The magnetic dipole source is placed at $(0, 0, 10R_c)$ and polarized with $(0, 1, 0)$.

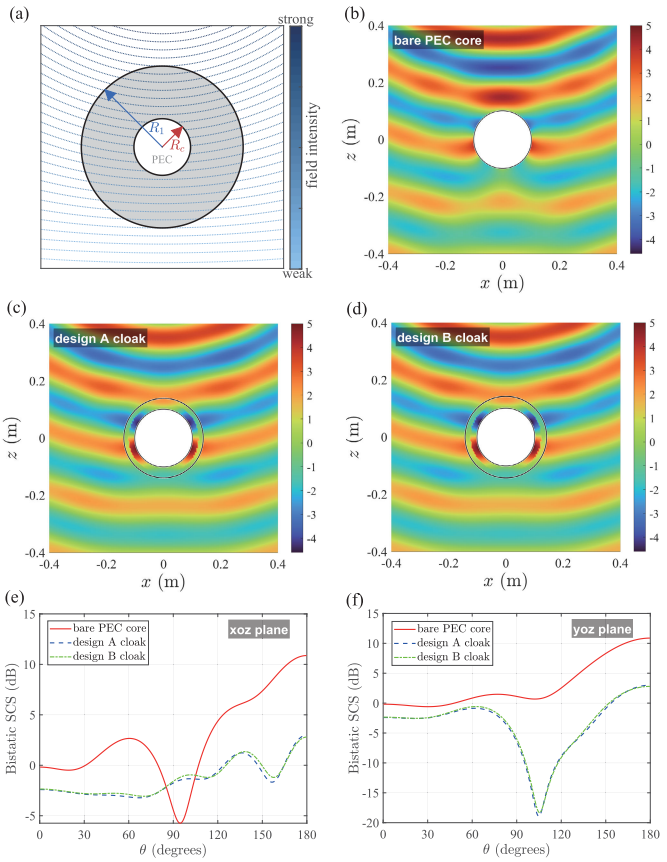


Fig. 11. Cloaking performance of the three-layer sphere shown in Fig. 10 computed by SIM. (a) Idealized perfect cloaking EM-field pattern. (b) Computed **E**-field pattern for the bare PEC core. (c) Computed **E**-field pattern for Design A. (d) Computed **E**-field pattern for Design B. (e) Comparisons of the bistatic SCS in the *xoz*- plane. (f) Comparisons of the bistatic SCS in the *yoz*- plane.

Fig. 11 shows the cloaking performance. Both the field distributions in the vicinity of the scatterer and the SCS values are computed using multilayered SIM. The EM-field pattern depicted in Fig. 11(a) demonstrates the anticipated ideal cloaking effect for a dipole source, characterized by the absence of distortion near the scatterer and a gradual transition to a parallel stripe pattern in the far-field. Obviously, the computed **E**-field pattern near a bare PEC sphere, as shown in Fig. 11(b), deviates significantly from the ideal scenario, exhibiting pronounced backscattering and stripe distortion. By

TABLE II

GEOMETRIC AND CONSTITUTIVE PARAMETERS OF THE THREE-LAYER EM CLOAKING STRUCTURE

Sphere \ Parameters	Design A			Design B		
	R/λ_0	ε	μ	R/λ_0	ε	μ
#1	$2.185/\pi$	3.480	0.500	$2.240/\pi$	3.030	0.520
#2	$1.935/\pi$	0.500	1.040	$1.940/\pi$	0.500	1.040
#3	0.500	–	–	0.500	–	–

Remark: spheres are numbered from the outermost one to the innermost one.

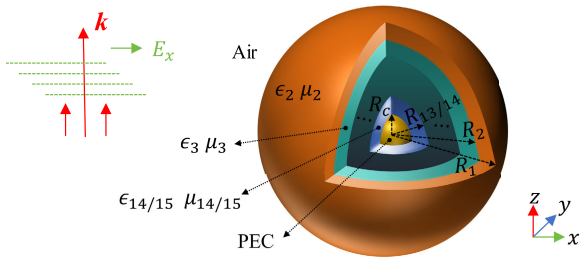


Fig. 12. Fourteen-layer or fifteen-layer cloaked sphere with a metal core and thirteen or fourteen dielectric shells illuminated by a plane wave.

contrast, when the metal core is surrounded by shells with parameters specified in Table II, the computed **E**-field patterns shown in Fig. 11(c) and (d) closely resemble the ideal pattern in Fig. 11(a). The dielectric shells effectively guide the electric fields around the PEC core, with only minor deviations from the ideal case, and the shadow region behind the metal sphere is eliminated. This superior cloaking performance is further corroborated by the bistatic SCS curves for both Designs A and B in the *xoz*- and *yoz*- planes, as shown in Fig. 11(e) and (f). These findings demonstrate that the proposed multilayered SIM can reliably and efficiently assess the EM cloaking effect when a limited number of dielectric spherical shells surrounding a metal core are excited by magnetic dipoles.

B. Metal Sphere Cloaked by Multiple Dielectric Layers Under Plane-Wave Illumination

In this case, we apply our multilayered SIM to more challenging and practical EM cloaking designs. As shown in Fig. 12, the configuration includes a significantly greater number of dielectric layers encasing the metal core compared to the designs discussed in Section IV-A. The incident fields are plane waves, and the operating frequency is 300 MHz. Two design types are assessed. The first design employs alternating thin layers of homogeneous materials, as proposed in [41], with parameters determined according to the formulas provided in Appendix. In this investigation, we specifically examine the design featuring 14 dielectric layers surrounding the metal sphere, with the radius of the outermost sphere being two times that of the metal core. The second design involves the optimization of constitutive parameters for 13 outer shells encasing the metal sphere, achieved through the application of particle swarm optimization in [42]. The parameters for these two design types are listed in Table III. We can see that a lot of thin layers are present in the design and the contrast of constitutive parameters is also very high in two adjacent layers.

Fig. 13 shows the cloaking performance. It is evident that the first design, which employs alternating thin layers, is

TABLE III

GEOMETRIC AND CONSTITUTIVE PARAMETERS OF THE FOURTEEN- AND FIFTEEN-LAYER EM CLOAKING STRUCTURES

Sphere \ Parameters	alternating cloak			optimized cloak		
	R/λ_0	ε	μ	R/λ_0	ε	μ
#1	1.966000	0.267949	0.267949	1.187177	7.227612	2.478084
#2	1.895786	3.752913	3.752913	1.171579	0.131510	0.110672
#3	1.825571	0.225760	0.225760	1.155981	0.498421	5.419352
#4	1.755357	3.795996	3.795996	1.140383	0.134066	0.199034
#5	1.685143	0.181881	0.181881	1.124785	9.971361	5.088651
#6	1.614929	3.840523	3.840523	1.109186	1.768592	3.209067
#7	1.544714	0.136918	0.136918	1.093588	0.294915	0.026139
#8	1.474500	3.885618	3.885618	1.077990	9.748927	3.311049
#9	1.404286	0.092122	0.092122	1.062392	4.096512	9.992937
#10	1.334071	3.929506	3.929506	1.046794	0.015038	9.985788
#11	1.263857	0.050008	0.050008	1.031196	9.677464	0.022260
#12	1.193643	3.968612	3.968612	1.015598	9.917844	4.652834
#13	1.123429	0.015687	0.015687	1.000000	0.008208	8.907839
#14	1.053214	3.995551	3.995551	0.983000	–	–
#15	0.983000	–	–	–	–	–

Remark: spheres are numbered from the outermost one to the innermost one.

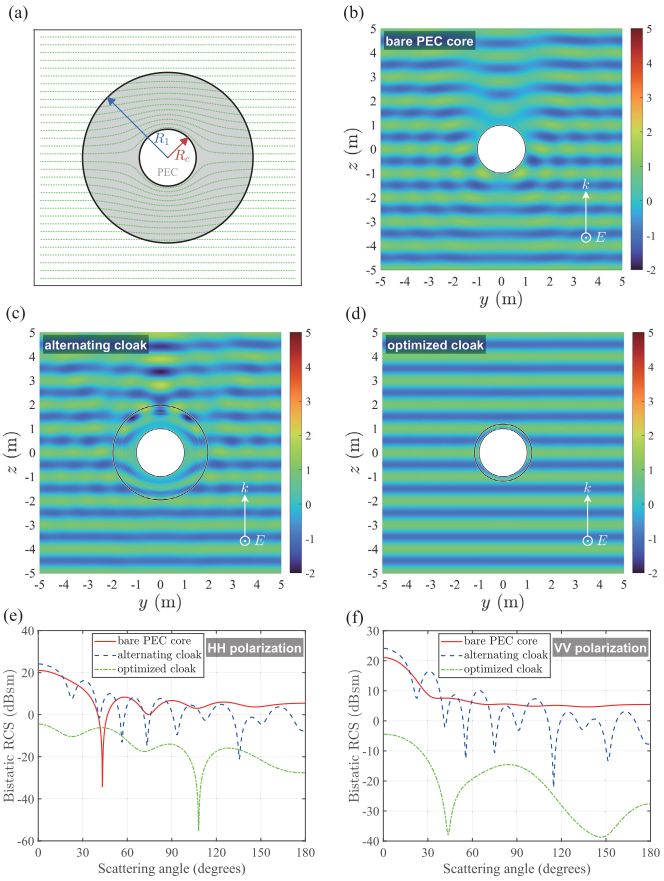


Fig. 13. Cloaking performance of the multilayered sphere shown in Fig. 12 computed by SIM. (a) Idealized perfect cloaking EM-field pattern. (b) Computed **E**-field pattern for the bare PEC core. (c) Computed **E**-field pattern for the alternating fifteen-layer cloak. (d) Computed **E**-field pattern for the optimized fourteen-layer design. (e) Comparisons of the bistatic RCS for HH polarization. (f) Comparisons of the bistatic RCS for VV polarization.

suboptimal. As shown in Fig. 13(c), the **E**-field pattern exhibits noticeable diffraction in the vicinity region of the sphere, which is slightly superior to the pure reflection observed in Fig. 13(b) for a bare PEC core. However, the pattern stripes are not straight when plane waves traverse the scatterer. According to the experiments presented in [41], the ideal cloaking performance can only be achieved with an infinite number

of alternating layers, which would substantially increase both design and computational costs. Fortunately, the ideal cloaking performance, characterized by straight pattern stripes on both the anterior and posterior sides of the scatterer, can be realized using the second design with optimized constitutive parameters determined by the particle swarm, as shown in Fig. 13(d). The significant differences in cloaking effects between the two designs are further substantiated by the bistatic RCS curves presented in Fig. 13(e) and (f). It is observed that the RCS values for the first design are only marginally lower than those for the EM scattering of a bare PEC sphere. In contrast, the RCS values for the second design are markedly lower than those for the bare PEC sphere. These computed results confirm that the proposed multilayered SIM can effectively evaluate the EM cloaking effect when a large number of dielectric spherical thin layers (several percentages of λ_0) surround a metal core and two adjacent layers have high dielectric contrast (ranging from several tens to over 1200 times).

V. Summary, Conclusion, and Future Work

In this work, we developed a multilayered spectral-domain PMCHWT-EFIE to analyze EM scattering from concentric spherical structures consisting of a metal core surrounded by an arbitrary number of dielectric layers. Dipole transmitters and receivers can be placed within any of these layers. This method extends the previously developed SIM applied to a single metal or dielectric sphere. The equivalent surface current densities on all spherical surfaces are expanded using the VSH series. These current densities are not only self-coupled with the EM fields on the same spherical surface through multiplication with the VSH expansions of DGFs, but they are also mutually coupled with the EM fields on adjacent spherical surfaces in a similar manner. The self-coupled spectral-IEs are discretized and incorporated into the diagonal submatrices of the final spectral-domain impedance matrix of PMCHWT-EFIE. In contrast, the mutually coupled equations are discretized and incorporated into the off-diagonal submatrices. The spectral-domain radiation matrix is constructed as a direct superposition of two radiation matrices, each assembled for one adjacent spherical boundary.

The accuracy and efficiency of the proposed multilayered SIM have been thoroughly validated through three numerical examples with distinct geometric scales and excitation configurations. The computed results have been cross-verified with those obtained from the commercial software COMSOL or Feko, as well as compared with calculations performed using the Mie series and the spatial-domain MoM. In one example, the results of multilayered SIM demonstrate excellent agreement with the Mie series solution while employing a notably low truncation order L. In the other two examples, the results of multilayered SIM demonstrate excellent agreement with reference solutions while necessitating significantly lower sampling densities, thereby reducing computational resources by several orders of magnitude compared to MoM. Furthermore, the practical applicability of the method has been demonstrated through the computation of two sets of EM cloaking designs, thereby affirming its potential applications in engineering designs.

Finally, it is important to highlight that, although the present study is confined to the canonical model, which includes only multilayered concentric standard spheres, the mathematical formulations and analyses presented herein constitute a solid step towards extending the technique to multilayered, arbitrarily shaped smooth structures, such as ellipsoids by invoking the Gaunt coefficient. Additionally, in certain practical applications, the innermost core is dielectric rather than metallic. The mathematical formulas derived in this work can be readily extended to account for EM scattering from multilayered spherical scatterers by replacing the EFIE with the PMCHWT equations for the innermost layer. Both $\bar{\mathbf{J}}$ and $\mathbf{M}$ are solved on the surface of the innermost sphere. The total number of unknowns in (5) and the dimensions of the spectral-domain impedance matrix $\mathbf{Z}$ in (9) become $4\Sigma_{n=1}^{N}\{(L_n+1)^2-1\}$. More challenging future work will involve extending SIM to the computation of EM scattering from 3-D inhomogeneous objects with non-smooth surfaces. Due to the inherent limitations of SIM, it cannot independently address this type of scattering problem. A viable approach is to integrate SIM with the SEM, where SEM addresses inhomogeneous regions and non-smooth surfaces, while SIM computes EM scattering from a fictitious smooth surface at a relatively low cost. This scheme has been successfully implemented in 2-D EM scattering [31], [32], [33], [34].

Appendix

Based on the design theory of EM cloak outlined in [43], the anisotropic parameters commonly employed for spherical structures, denoted as $\rho_0\mathrm{diag}\{\rho_r,\rho_\theta,\rho_\phi\}$, can be equivalently represented by two alternating isotropic parameters, ρ_A^n and ρ_B^n. They are related to the anisotropic parameters by

$$\rho_r=\frac{2}{\rho_A^{n-1}+\rho_B^{n-1}},\quad \rho_\theta=\rho_\phi=\frac{\rho_A^n+\rho_B^n}{2} \tag{A1}$$

where ρ can be ε or μ, and n is the layer index. We take the alternating cloak with 14 dielectric shells described in Section IV-B as an example to derive their isotropic parameters. The radius of the innermost metal sphere is R_c, while the outermost radius of the cloak is $R_1=2R_c$. Consequently, each shell possesses a thickness of $\Delta=R_c/14$ and its radius is

$$R_c<R_n=R_c+(15-n)\,\Delta\le 2R_c,\ n\in[1,14]\,. \tag{A2}$$

According to the effective medium theory, when the shell thickness is uniform and sufficiently small, the anisotropic parameters of the spherical cloak are proposed as [41]

$$\begin{cases}\rho_r=\left(\dfrac{2R_c}{2R_c-R_c}\right)\left(\dfrac{R_n-R_c}{R_n}\right)^2=2\left(\dfrac{R_n-R_c}{R_n}\right)^2\\ \rho_\theta=\rho_\phi=\dfrac{2R_c}{2R_c-R_c}=2.\end{cases} \tag{A3}$$

Combining (A1) and (A3) directly yields

$$\begin{cases}\rho_A^n=2-2\sqrt{\left(2-\dfrac{R_c}{R_n}\right)\dfrac{R_c}{R_n}},\quad n=2k-1,\ k\in[1,7]\\ \rho_B^n=2+2\sqrt{\left(2-\dfrac{R_c}{R_n}\right)\dfrac{R_c}{R_n}},\quad n=2k,\ k\in[1,7]\end{cases} \tag{A4}$$

where ρ_A represents the constitutive parameters of spherical shells with odd indices, while ρ_B represents those with even indices, from the outermost shell to the innermost one.

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